UNIVERSITÉ CÔTE D'AZUR

Système de Tracking

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Année-académique : 2025-2026

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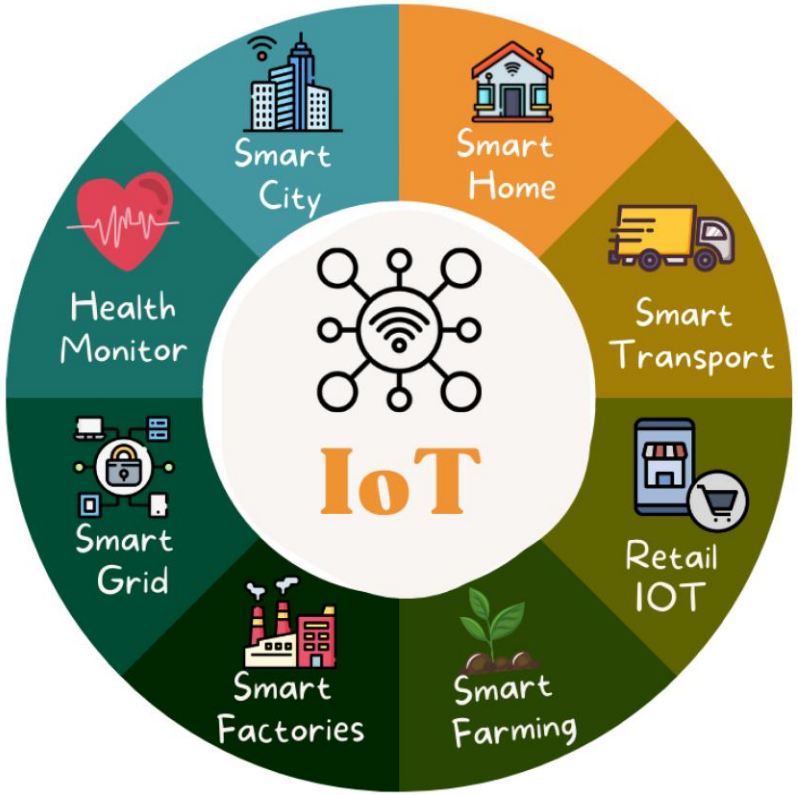
I. Introduction

- **Croissance IoT**

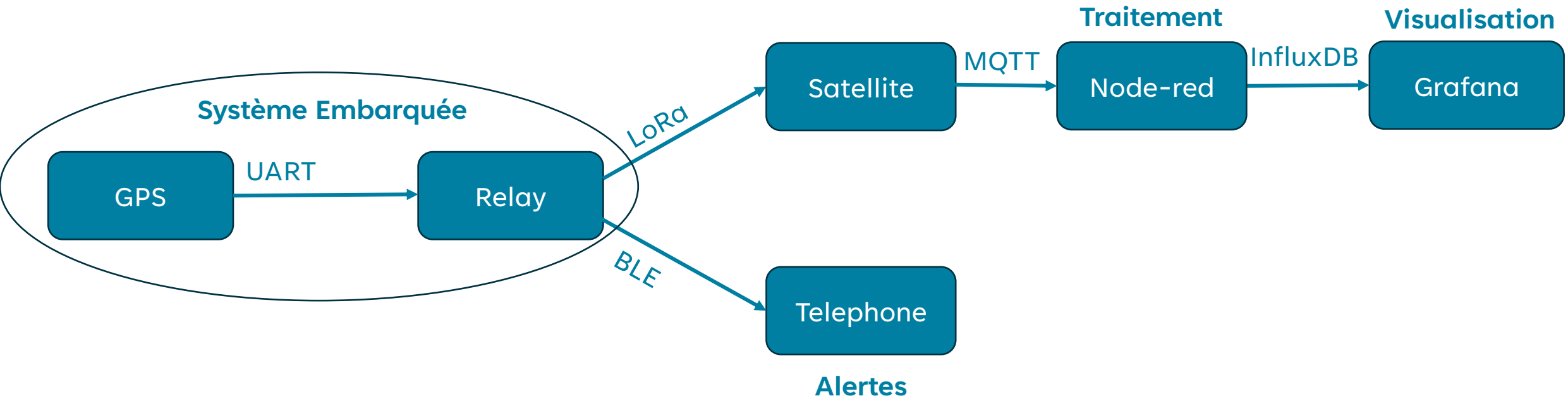


- autonomie
- portée
- fiabilité

Système de Tracking

- Suivi de position en temps réel
- Transmission longue portée via LoRa
- Alertes locales via BLE
- Optimisation de la consommation énergétique

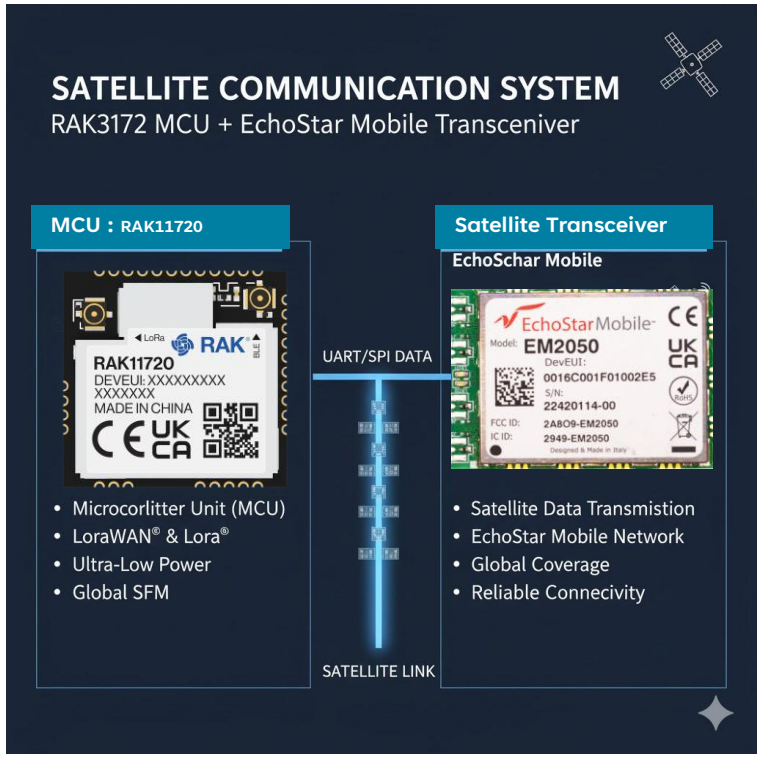
II. Architecture du Système

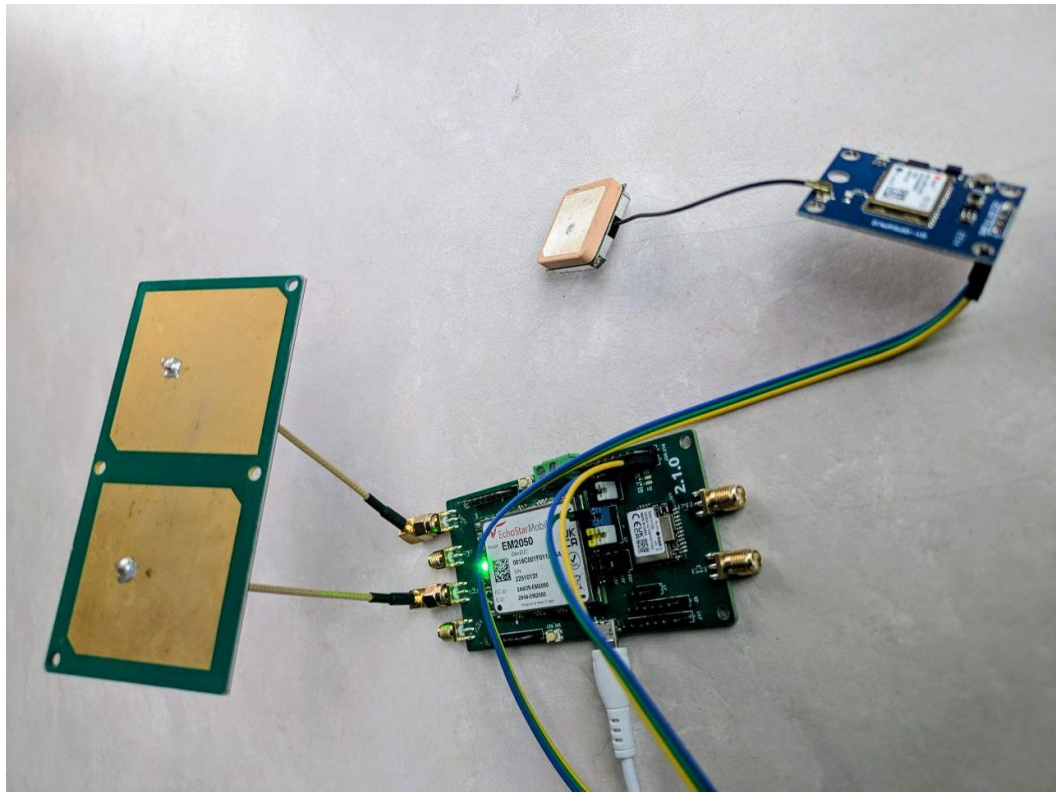


III. Système Embarquée

1. GPS

2. Relay





III. Système Embarquée

3. Implémentation Logiciel

Zephyr RTOS

Threads :

- ❑ **GPS**
- ❑ **BLE**
- ❑ **LoRa**

```
LAT=43.696826 LON=7.247246
LAT=4341.80960 N  LON=00714.83463 E

LAT=43.696827 LON=7.247244
LAT=4341.80951 N  LON=00714.83477 E

LAT=43.696825 LON=7.247246
LAT=4341.80948 N  LON=00714.83479 E

LAT=43.696825 LON=7.247247
LAT=4341.80942 N  LON=00714.83486 E

LAT=43.696824 LON=7.247248
UART1 ready = 1
LAT=4341.80935 N  LON=00714.83493 E

LAT=43.696823 LON=7.247249
LAT=4341.80933 N  LON=00714.83489 E

LAT=43.696822 LON=7.247248
LoRa TX: AT+SEND=1,0,8,1,006E958F029AC2B7
LAT=4341.80924 N  LON=00714.83497 E

LAT=43.696821 LON=7.247249
LAT=4341.80915 N  LON=00714.83506 E

LAT=43.696819 LON=7.247251
BLE Sent -> 0 Km parcouru !!!
LAT=4341.80909 N  LON=00714.83508 E

LAT=43.696818 LON=7.247251
LAT=4341.80901 N  LON=00714.83515 E

LAT=43.696817 LON=7.247253
LAT=4341.80900 N  LON=00714.83510 E
```

IV. Transmission LoRa & BLE

LoRa

- ❑ Transmission **longue portée**
- ❑ Transmission de la **position**
- ❑ Permet **de retracer les trajets**

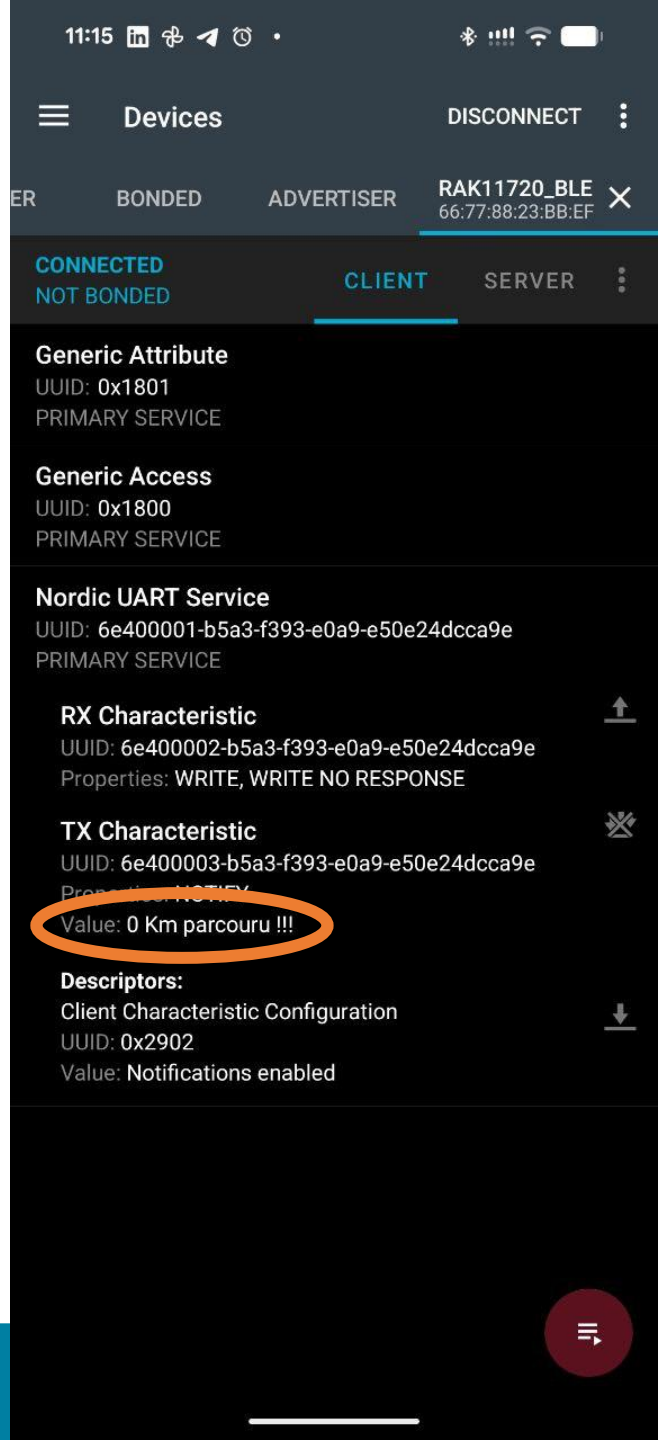
BLE

- ❑ Transmission **courte portée**
- ❑ Transmission de la **distance**
- ❑ Emission des **alertes**

Payload envoyé :

8 octets

- ❑ 4 bytes latitude
- ❑ 4 bytes longitude



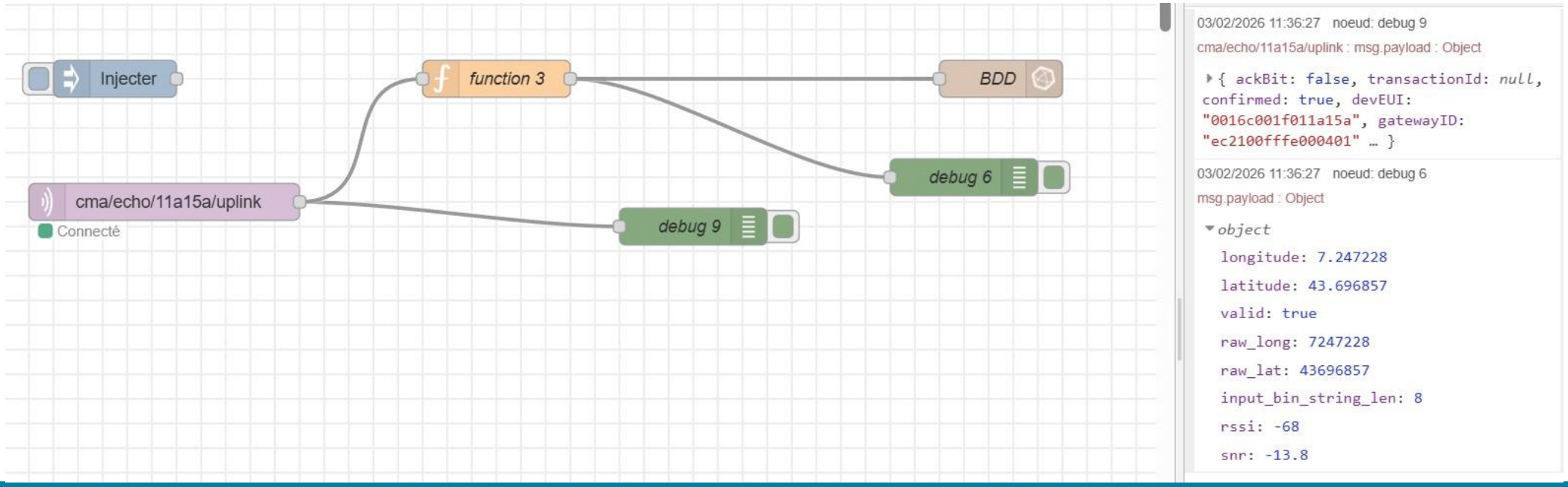
V. Traitement Node-Red

MQTT

- ❑ Protocole léger publish
- ❑ Adapté IoT
- ❑ Gestion du flux en temps réel

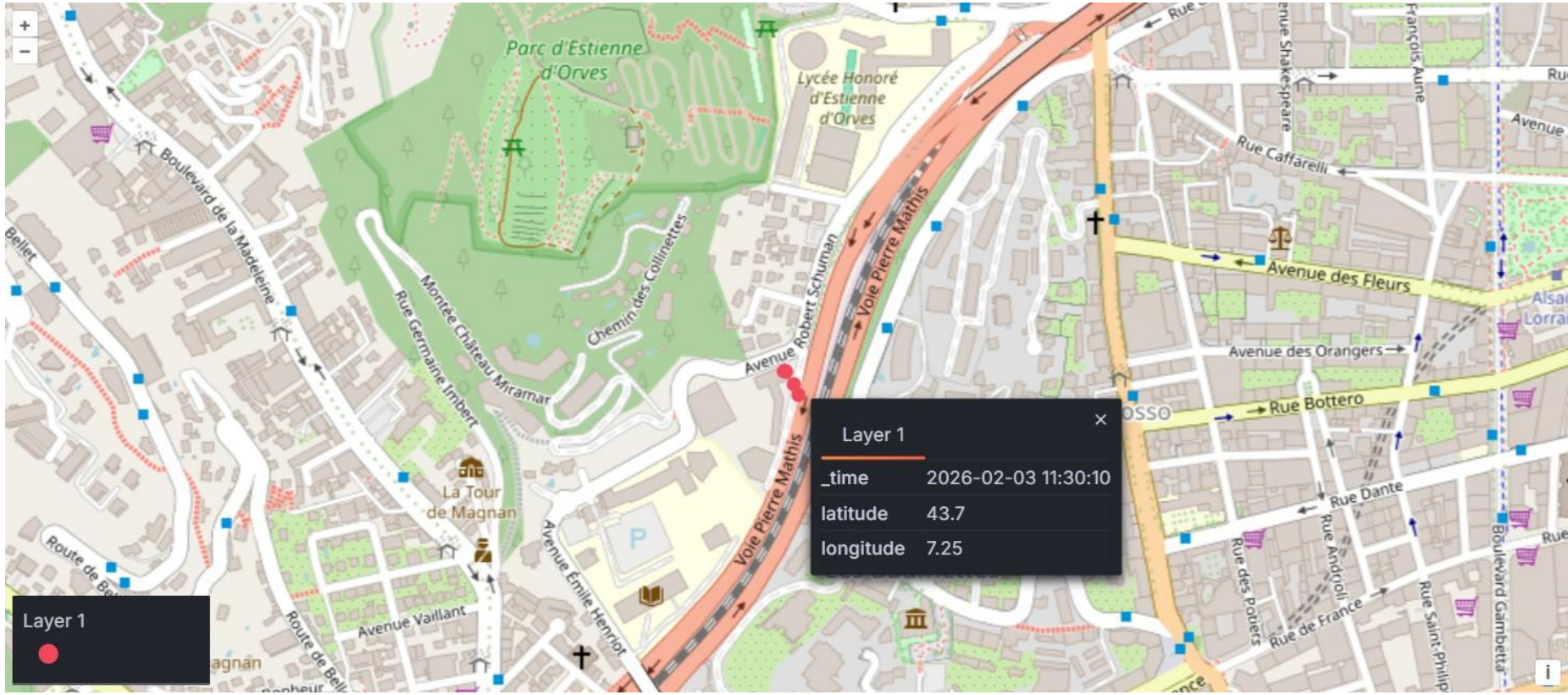
Node-RED

- ❑ Décodage du payload (latitude, longitude)
- ❑ Formatage des données en structure exploitable
- ❑ Redirection vers la base de données InfluxDB



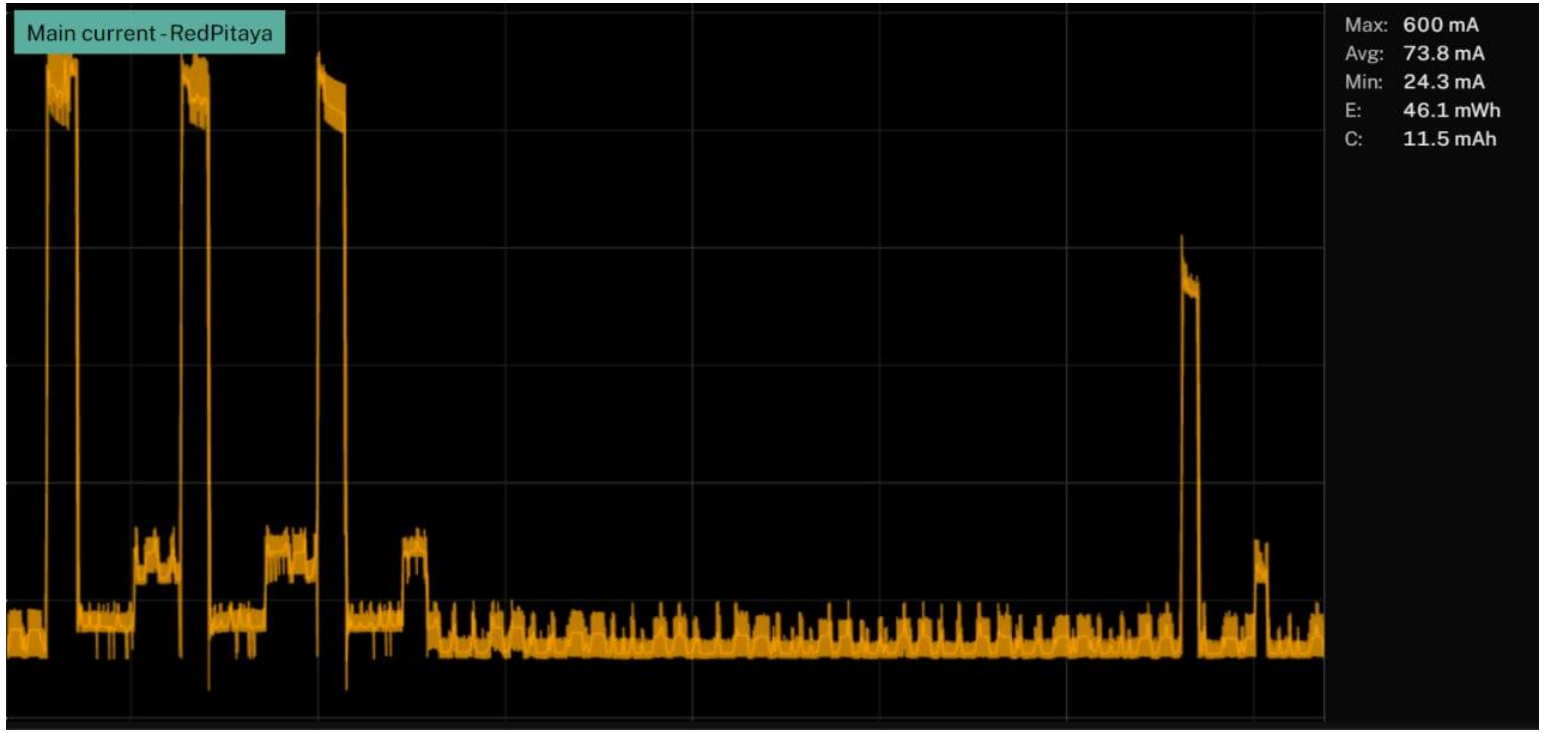
VI. Visualisation Grafana

- Connexion directe à **InfluxDB**
- Affichage des coordonnées **GPS**
- Visualisation **du trajet sur carte**
- Suivi en **temps réel** de la position



VII. Gestion énergétique

- **532 mA** during **join** mode for **1.478 s**
- **63.4 mA** when **waiting** for **4 min**
- **308 mA** during **transmission** mode for **914 ms**



VIII. Difficultés rencontrées

- Conflit radio **BLE / LoRa**
- Gestion **RTOS**
- **Encodage des données**
- **Channel Busy**
- Synchronisation **radio**

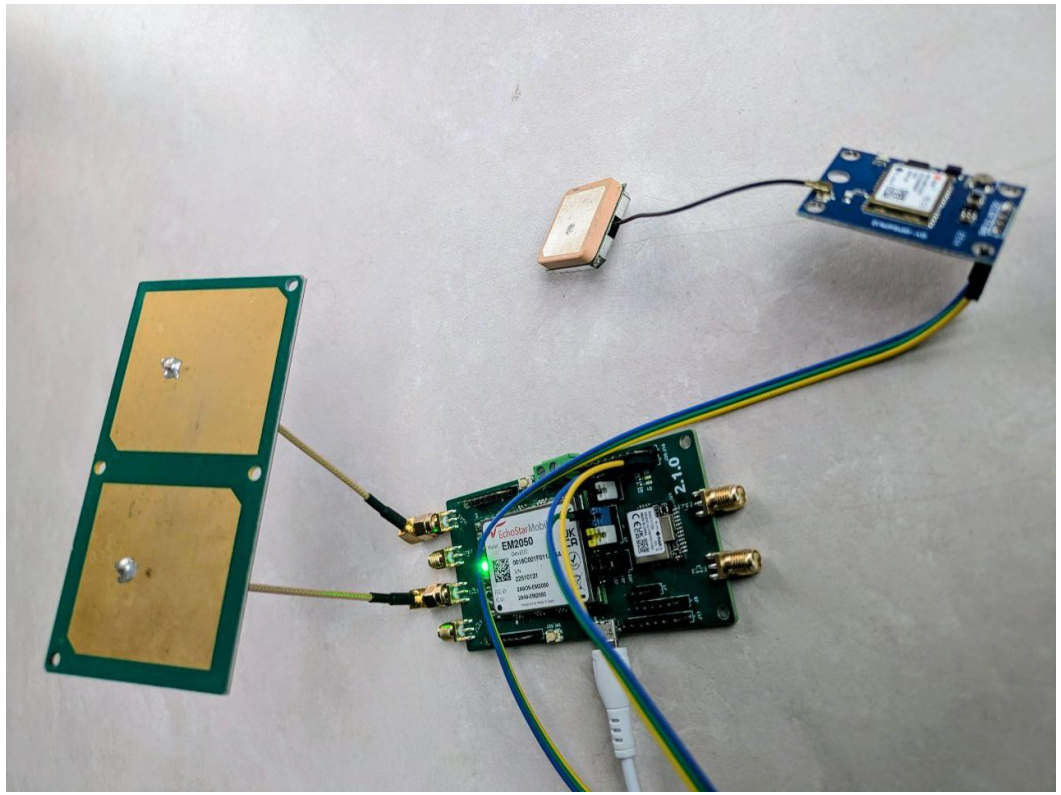
IX. Conclusion & Perspectives

Résultat

- ❑ Chaîne IoT fonctionnelle
- ❑ Transmission stable
- ❑ Visualisation temps réel

Améliorations possibles

- ❑ Optimisation consommation
- ❑ Ajout capteurs
- ❑ Application mobile dédiée
- ❑ Ajout d'un piezo pour émettre des vibrations au moment de la transmission d'alertes



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THANK YOU FOR YOUR ATTENTION

Feel free to ask any question